Math-2205 Class Test 02 Section E

1. Consider the table below for the followings.

x	19	25	23	22
y	29	23	26	27

(i). Find the correlation coefficient. **[2]**
(ii). Find regression line $\hat{x} = a + by$. **[4]**
(iii). Estimate the missing value in the ordered pair $(?,25)$. **[1]**

2. Explain the correlation coefficient $r = -0.8$ and roughly sketch the corresponding scatter diagram. Also, find the coefficient of determination. **[3]**

Math-2205 Class Test 02 Section G

1. Explain the terms $r^2 = 0.81$ and $b_{y/x} = -2.31$. **[2]**

2. Consider the correlation coefficient of two variables is 0.65 and regression coefficient of y on x is 0.92. Also, $\bar{x} = 12.6$ and $\bar{y} = 25.1$. Then, find and manifest the regression line of x on y. **[5]**

3. Find the rank correlation co-efficient between obtained places of 5 candidates in two programming contests. **[2]**

Contest 1	3	1	4	5	2
Contest 2	4	3	1	2	5

4. Sketch the rough scatter diagram for $r = -0.75$. **[1]**

Math-2205 Class Test 02 Section O

1. Consider the table below for the followings.

x	19	25	23	22
y	29	23	26	27

(i). Find the coefficients $b{y}/_{x}$ and $b{x}/_{y}$ and hence estimate r_{xy}. **[4]**

(ii). Which of x and y are dominant? Why? **[1]**

(iii). Explain the correlation coefficient r_{xy} and roughly sketch the corresponding scatter diagram. **[2]**

2. What you understand by $r^2 = 0.35$? [**1]**

3. Find the rank correlation co-efficient between obtained places of 5 children in two sporting events. **[2]**

Event 1	3	4	5	2	1
Event 2	3	1	2	5	4

Math-2205 Class Test 02 Section P

1. If the correlation coefficient of $x \,\&\, y$ is 0.85 and the corresponding standard deviations 1.52 & 2.31.
   (i). Find and explain the regression coefficient of y on x. **[2]**
   (ii). For $\bar{x} = 27.1$ and $\bar{y} = 15.3$ find and sketch the regression line of y on x. **[4]**
   (iii). Estimate the missing value in the ordered pair $(30, ?)$. **[1]**

2. According to following scatter disgram, assume what can be the value of the coefficient of regression and explain the situation by sketching a rough regression line within the provided data. **[3]**